

Databases - Laboratory Work 1

ER Diagram: International Airport Database

This document contains the ER diagram of the International Airport database and a textual description of the solution, according to the tasks of Laboratory Work 1.

1. Identified Entities

1. Airport - an airport that can serve as a departure or arrival point for flights.
2. Airline - an airline company that operates flights.
3. Flight - a scheduled or actual flight operated by an airline between two airports.
4. Passenger - a person who books flights.
5. Booking - a reservation made by a passenger for a specific flight.
6. BookingChange - a record of any change made to an existing booking.
7. BoardingPass - a boarding pass issued for a booking.
8. Baggage - a piece of baggage registered for a booking.
9. BaggageCheck - the result of checking a piece of registered baggage.
10. SecurityCheck - the result of a passenger's security screening.

2. Entity Attributes

Primary keys are marked as PK, foreign keys are marked as FK.

Airport:

Attribute	Type	Key
airport_id	INT	PK
airport_name	VARCHAR	
country	VARCHAR	
state	VARCHAR	
city	VARCHAR	
created_at	DATETIME	
updated_at	DATETIME	

Airline:

Attribute	Type	Key
airline_id	INT	PK

airline_code	VARCHAR	
name	VARCHAR	
country	VARCHAR	
created_at	DATETIME	
updated_at	DATETIME	

Flight:

Attribute	Type	Key
flight_id	INT	PK
departing_gate	VARCHAR	
arriving_gate	VARCHAR	
airline_id	INT	FK -> Airline
departure_airport_id	INT	FK -> Airport
arrival_airport_id	INT	FK -> Airport
scheduled_departure_time	DATETIME	
scheduled_arrival_time	DATETIME	
actual_departure_time	DATETIME	
actual_arrival_time	DATETIME	
created_at	DATETIME	
updated_at	DATETIME	

Passenger:

Attribute	Type	Key
passenger_id	INT	PK
first_name	VARCHAR	
last_name	VARCHAR	
gender	VARCHAR	
date_of_birth	DATE	
country_of_citizenship	VARCHAR	
country_of_residence	VARCHAR	
passport_number	VARCHAR	

created_at	DATETIME	
updated_at	DATETIME	

Booking:

Attribute	Type	Key
booking_id	INT	PK
flight_id	INT	FK -> Flight
passenger_id	INT	FK -> Passenger
status	VARCHAR	
booking_platform	VARCHAR	
ticket_price	DECIMAL	
created_at	DATETIME	
updated_at	DATETIME	

BookingChange:

Attribute	Type	Key
change_id	INT	PK
booking_id	INT	FK -> Booking
change_description	VARCHAR	
created_at	DATETIME	
updated_at	DATETIME	

BoardingPass:

Attribute	Type	Key
boarding_pass_id	INT	PK
booking_id	INT	FK -> Booking
seat	VARCHAR	
boarding_time	DATETIME	
created_at	DATETIME	
updated_at	DATETIME	

Baggage:

Attribute	Type	Key
baggage_id	INT	PK
booking_id	INT	FK -> Booking
weight_kg	DECIMAL	
created_at	DATETIME	
updated_at	DATETIME	

BaggageCheck:

Attribute	Type	Key
baggage_checking_id	INT	PK
booking_id	INT	FK -> Booking
passenger_id	INT	FK -> Passenger
check_results	VARCHAR	
created_at	DATETIME	
updated_at	DATETIME	

SecurityCheck:

Attribute	Type	Key
security_check_id	INT	PK
passenger_id	INT	FK -> Passenger
check_results	VARCHAR	
created_at	DATETIME	
updated_at	DATETIME	

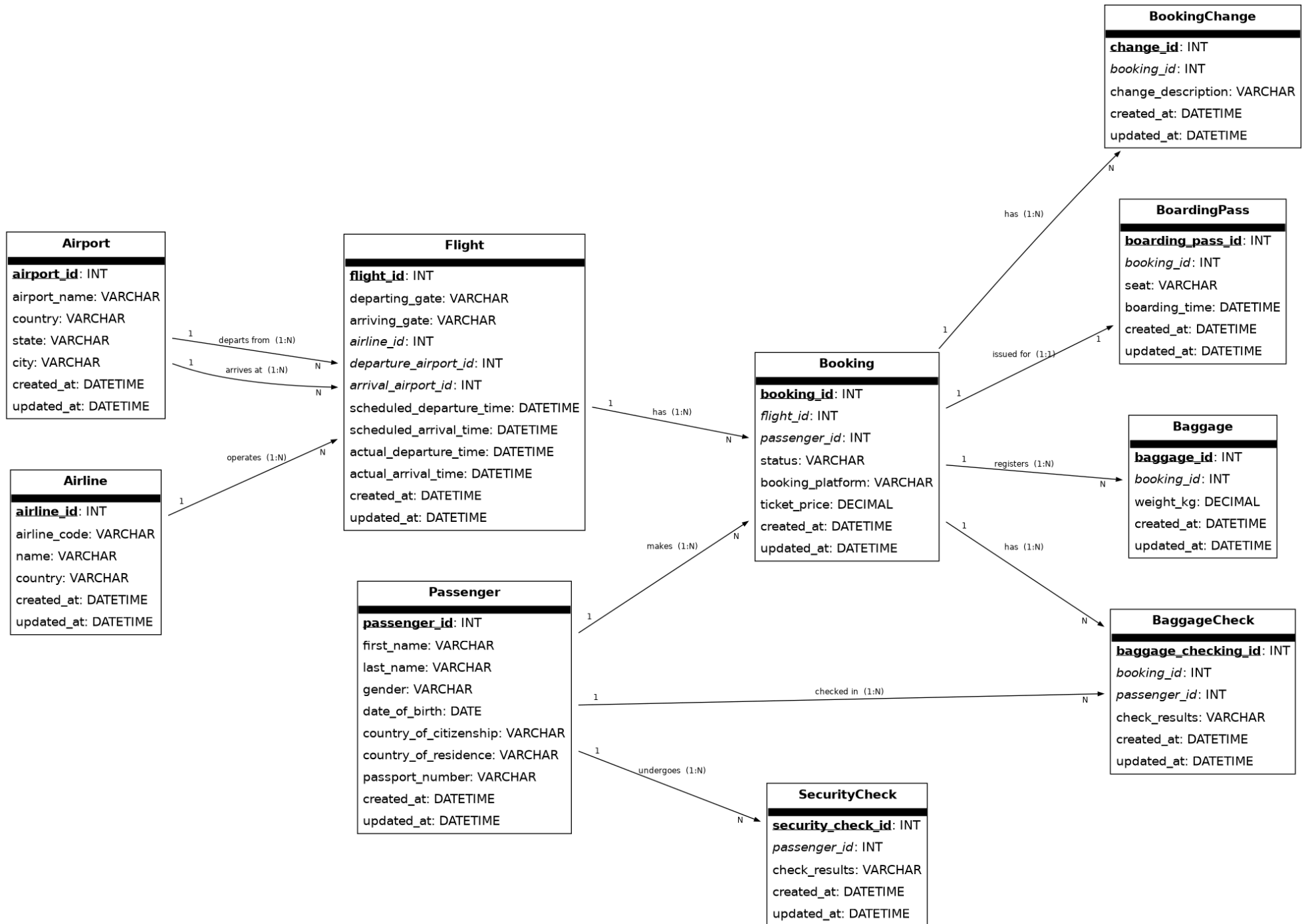
3. Relationships and Cardinality

Relationship	Cardinality	Description
Airport - Flight	1:N	one airport is the departure point for many flights
Airport - Flight	1:N	one airport is the arrival point for many flights
Airline - Flight	1:N	one airline operates many flights

Flight - Booking	1:N	one flight can have many bookings
Passenger - Booking	1:N	one passenger can make many bookings
Booking - BookingChange	1:N	one booking can have many recorded changes
Booking - BoardingPass	1:1	each booking is issued exactly one boarding pass
Booking - Baggage	1:N	one booking can register many pieces of baggage
Booking - BaggageCheck	1:N	one booking can have many baggage checks
Passenger - BaggageCheck	1:N	one passenger's baggage can be checked multiple times
Passenger - SecurityCheck	1:N	one passenger can undergo multiple security checks

All relationships are one-to-many (1:N), except Booking - BoardingPass which is one-to-one (1:1). There are no direct many-to-many relationships; the Booking entity works as the link between Passenger and Flight.

4-6. ER Diagram



7. Legend and Notation

- Each rectangle represents one entity (a database table); the header of the rectangle shows the entity name.
- Attributes are listed inside the entity box, one per line, in the format name (TYPE).
- PK next to an attribute means Primary Key - it uniquely identifies each record.
- FK next to an attribute means Foreign Key - it references the primary key of another entity.
- Lines between entities represent relationships; the numbers near each end of the line (1 or N) show the cardinality of that side.
- All relationships are mandatory on the many side (for example, a Booking cannot exist without a Flight and a Passenger).

Summary of the Solution

The Airport database is centred on the Booking entity, which links a Passenger to a Flight. A Flight references the operating Airline and two Airport records, one for departure and one for arrival. Each booking may generate a BoardingPass (one per booking), any number of BookingChange entries if the booking is modified, and any number of registered Baggage items, each later verified through a BaggageCheck. Separately, a Passenger also undergoes a SecurityCheck before boarding. Historical data such as changes and checks is kept in separate tables rather than overwriting values, and every table uses a surrogate integer primary key together with created_at and updated_at audit columns.